

Progress Report

Depth of Field Research

Rui Gao — Sep 13

Week 3 · Sep 7 – Sep 13, 2026

Progress

- Worked through the OpenGL graphics pipeline as a series of numbered experiments, each saved with code and notes.
- Vertex attributes and interpolation (VBO/VAO), indexed drawing (EBO), and uniforms with animation.
- Transformation matrices, the first 3D cube, depth testing, and model-view-projection matrices.
- Built an interactive first-person camera (WASD movement and mouse look).
- Visualized the raw depth buffer and reconstructed linear view depth, the quantity a DoF effect needs.

Meeting — Sep 8

Attendees: Rui Gao, Beibarys Myrzash, Professor Lingqi Yan.

- Presented first progress: the first rendered images, and 2D and 3D scenes.
- Asked questions about the final AI solution.
- Received next tasks: research framebuffers and current DoF methods, render with multiple passes in OpenGL, and write a blur shader.

Next Steps

- Render the scene off-screen into a framebuffer with color and depth textures.
- Compute the circle of confusion from depth and write a first blur shader.